HYPER TEXT MARKUP LANGUAGE

WHAT IS HTML?

- **HyperText Markup Language (HTML)** is the standard markup language used to create and design the structure of web pages.
- It provides the basic building blocks of a webpage and organizes content into a structured format using elements, tags, and attributes.

KEY FEATURES OF HTML

- **Platform Independent:** Works seamlessly across all browsers and operating systems.
- **Flexible:** Compatible with additional technologies like CSS for styling and JavaScript for interactivity.
- **Linking Capability:** Hyperlinks (anchors) can connect to other web pages or resources.
- **Media Support:** Embeds multimedia like images, videos, and audio files.
- **Lightweight and Fast:** Efficient for creating static and responsive pages.

BASIC STRUCTURE OF AN HTML DOCUMENT

- <!DOCTYPE html>
- <html>
- <head>
- <title>Title of the Page</title>
- </head>
- <body>
- <h1>Welcome to HTML</h1>
- This is a sample HTML document.
- </body>
- </html>

BASIC STRUCTURE OF AN HTML DOCUMENT

- <!DOCTYPE html>: Defines the document type and version of HTML (HTML5 is the latest).
- <html>: Root element containing all HTML code.
- <head>: Holds metadata, including titles, styles, and linked resources.
- <title>: Displays the page title in the browser tab.
- <body>: Contains visible page content, such as text, images, and multimedia.

HTML SEMANTIC TAGS

- **Definition:**
  Semantic tags are HTML5 elements that clearly define the role and purpose of the content they encapsulate.
- These tags enhance the readability and accessibility of HTML documents, providing additional meaning to web pages for browsers, developers, and assistive technologies.

HTML SEMANTIC TAGS

- **Benefits of Using Semantic Tags**
  - **Improved Code Readability:** Makes the code easier for developers to understand.
  - **SEO Benefits:** Search engines can better interpret the structure and content of the webpage.
  - **Accessibility:** Assistive devices like screen readers can better navigate and interact with the content.
  - **Future Proofing:** Complies with modern web standards and improves compatibility with evolving technologies.

HTML SEMANTIC/BASIC TAGS

Tag	Purpose	Example Usage
<!DOCTYPE>	Specifies the document type and version.	<!DOCTYPE html> (HTML5 declaration)
<html>	Defines the root of the HTML document.	<html> ... </html>
<head>	Contains metadata about the document.	<head> ... </head>
<title>	Sets the title of the webpage (displayed in browser tab).	<title>My Page</title>
<body>	Defines the main content of the document.	<body> ... </body>
<h1> to <h6>	Represents headings (h1 largest, h6 smallest).	<h1>Heading 1</h1>
	Denotes a paragraph of text.	This is a paragraph.
<a>	Creates hyperlinks.	<a href="https://example.com">Visit Site</a>
<img>	Embeds an image.	<img src="image.jpg" alt="Description">
 	Inserts a line break.	First Line Second Line
<hr>	Creates a horizontal rule (divider).	<hr>

HTML SEMANTIC/BASIC TAGS

Semantic Tag	Purpose	Example Usage
<header>	Represents introductory content or navigation links for a webpage or a section.	Wraps titles, logos, or introductory elements.
<footer>	Represents the footer section, typically for copyright notices or links.	Footer with social media links or site information.
<article>	Defines independent, self-contained content, like news articles or blog posts.	An individual blog post or article.
<section>	Groups related content under a specific thematic area.	A section of related items like "Features" or "Services".
<nav>	Represents a block of navigation links.	Navigation menus or site maps.
<aside>	Represents content tangentially related to the main content, like sidebars.	Ads, quotes, or related links displayed in the sidebar.
<main>	Represents the main content of the document, unique and central to the page.	The primary article or content area.
<figure>	Groups media elements with captions.	Images, illustrations, or charts with <figcaption>.
<figcaption>	Provides captions for the <figure> element.	Descriptive text for an image or graph.
<mark>	Highlights text that is relevant or important.	Highlighting important words in an article.
<time>	Represents a specific time or date.	Timestamp for a blog or an event.

<HEADER> EXAMPLE

- Represents the top section of the webpage with a title and navigation.
  - <header>
  - <h1>My Portfolio</h1>
  - <nav>
  - <ul>
  - <li><a href="#about">About</a></li>
  - <li><a href="#projects">Projects</a></li>
  - <li><a href="#contact">Contact</a></li>
  - </ul>
  - </nav>
  - </header>

<SECTION> EXAMPLE

- Groups related content under a theme.
  - <section id="features">
  - <h2>Features</h2>
  - Our application offers seamless integration with your favorite tools.
  - </section>

<ARTICLE> EXAMPLE

- Self-contained content like a blog post or news article.
  - <article>
  - <h2>The Importance of Web Standards</h2>
  - Web standards help ensure consistency and accessibility in web development...
  - </article>

<FIGURE> AND <FIGCAPTION> EXAMPLE

- Displays images or diagrams with captions.
  - <figure>
  - <img src="diagram.jpg" alt="A web development diagram">
  - <figcaption>Figure 1: The relationship between HTML, CSS, and JavaScript</figcaption>
  - </figure>

<FOOTER> EXAMPLE

- Footer section with copyright and contact information.
  - <footer>
  - © 2024 MyWebsite. All Rights Reserved.
  - Contact us: <a href="mailto:info@mywebsite.com">info@mywebsite.com</a>
  - </footer>

<ASIDE> EXAMPLE

- Sidebar content with additional information or advertisements.
  - <aside>
  - <h2>Sponsored</h2>
  - Check out this great deal at <a href="http://example.com">Example.com</a>!
  - </aside>

<MAIN> EXAMPLE

- Central content of the webpage.
  - <main>
  - <h2>Welcome to My Website</h2>
  - Here, you'll find tutorials, blogs, and resources about web development.
  - </main>

A SIMPLE HTML PAGE USING BASIC TAGS

```
<!DOCTYPE html>
<html>
<head>
    <title>Basic HTML Page</title>
</head>
<body>
    <h1>Welcome to HTML</h1>
    <p>This is a sample web page using basic HTML tags.</p>
    <a href="https://example.com">Click here to visit Example</a>
    <br>
    <img src="sample.jpg" alt="Sample Image">
    <hr>
</body>
</html>
```

HTML FORMATTING TAGS

- Formatting tags are used to style or emphasize specific parts of content, such as making text bold, italicized, or underlined.

Key Formatting Tags

Tag	Description	Example Usage
	Makes the text **bold**.	Bold Text
	Makes the text *italicized*.	Italic Text
<u>	Underlines the text.	<u>Underlined Text</u>
	Indicates strong emphasis (rendered as bold).	Important!
	Indicates emphasis (rendered as italicized).	Important Note
<mark>	Highlights text.	<mark>Highlighted Text</mark>
<small>	Makes text smaller than the surrounding text.	<small>Disclaimer</small>
<sub>	Displays text as subscript.	H₂O
<sup>	Displays text as superscript.	X²
<del>	Strikes through text to indicate deletion.	<del>Old Text</del>
<ins>	Underlines text to indicate insertion.	<ins>Added Text</ins>
<blockquote>	Adds an indented block of quoted text.	<blockquote>This is a quote.</blockquote>

EXAMPLE: USING FORMATTING TAGS

```
<!DOCTYPE html>
<html>
<head>
    <title>HTML Formatting Tags</title>
</head>
<body>
    <h1>Text Formatting Examples</h1>
    <p>This is a <b>bold</b> and <i>italicized</i> word.</p>
    <p>You can also <u>underline</u> text for emphasis.</p>
    <p>Water's chemical formula is: H<sub>2</sub>O</p>
    <p>Power of X is expressed as: X<sup>2</sup></p>
    <p><mark>Highlight</mark> important points in text.</p>
    <blockquote>This is a blockquote for longer quotations.</blockquote>
</body>
</html>
```

ADDING IMAGES TO A WEB PAGE

- Images are an integral part of web design, enhancing visual appeal and conveying information effectively.
- HTML provides the <img> tag for embedding images in web pages.

THE <IMG> TAG

- The <img> tag is an empty tag (no closing tag) used to embed images in HTML. Images can be linked from external sources or hosted locally.
- Basic Syntax:
  - <img src="URL or file path" alt="Description of the image">

ATTRIBUTES OF THE <IMG> TAG

Attribute	Description	Example
src	Specifies the source of the image (URL or file path).	<img src="image.jpg">
alt	Defines alternative text for accessibility if the image cannot load.	<img alt="Sample Image">
width	Specifies the width of the image in pixels or percentage.	<img width="300">
height	Specifies the height of the image in pixels or percentage.	<img height="200">
title	Provides a tooltip when hovering over the image.	<img title="Hover text">
loading	Determines loading behavior (lazy or eager).	<img loading="lazy">

ADDING AN IMAGE EXAMPLE

```
<!DOCTYPE html>
<html>
<head>
    <title>Adding Images</title>
</head>
<body>
    <h1>HTML Image Example</h1>
    <p>Below is an example of an embedded image:</p>
    <img src="sample.jpg" alt="A beautiful scenery" width="500" height="300" title="Scenery Image">
</body>
</html>
```

LINKING EXTERNAL IMAGES

- <img src="https://example.com/images/sample.jpg" alt="Example Image">

IMAGE AS A LINK

- You can wrap the <img> tag with the <a> tag to make the image clickable.
- <a href="https://example.com">
- <img src="sample.jpg" alt="Clickable Image" width="300">
- </a>

ADDING BACKGROUND IMAGES (USING CSS)

- While the <img> tag displays inline images, background images can be applied using CSS for decorative purposes.
- HTML:
  - This is a background image.
- CSS:
  - .image-container {
  - background-image: url('background.jpg');
  - height: 300px;
  - width: 500px;
  - color: white;
  - text-align: center;
  - line-height: 300px;
  - background-size: cover;
  - }

RESPONSIVE IMAGES

- Responsive images adjust to fit their container size, enhancing usability on different devices.
  - <img src="sample.jpg" alt="Responsive Image" style="max-width: 100%; height: auto;">

LAZY LOADING IMAGES

- Lazy loading defers the loading of images until they appear in the user's viewport, improving page performance.
  - <img src="sample.jpg" alt="Lazy Loaded Image" loading="lazy">

BEST PRACTICES FOR ADDING IMAGES

- Optimize Image Sizes: Compress images to reduce file size and improve load times.
- Use Descriptive alt Text: Helps with SEO and accessibility.
- Use Scalable Formats: Prefer formats like SVG or WebP for modern web design.
- Fallback Handling: Always include alt attributes to describe content in case of loading errors.
- Responsive Design: Ensure images adapt to various screen sizes using CSS or media queries.

LISTS

- Lists are a crucial part of HTML, helping organize and display data or content in an ordered or unordered format.
- HTML supports different types of lists to suit various needs, including bullet points, numbered steps, or descriptive formats.

TYPES OF LISTS

Type	Description	Tag
Unordered List	Displays items as bullet points.	<ul>
Ordered List	Displays items as a numbered or alphabetized sequence.	<ol>
Definition List	Displays items as terms and their descriptions.	<dl>, <dt>, <dd>

UNORDERED LIST (<UL>)

- Unordered lists display items with bullet points by default.
  - <ul>
  - <li>Item 1</li>
  - <li>Item 2</li>
  - <li>Item 3</li>
  - </ul>

CUSTOMIZATION WITH CSS

- You can customize bullet styles using the list-style-type property in CSS.
- Examples of list-style-type Values:

Value	Bullet Style	Example Code
`disc` (default)	•	`<ul style="list-style-type: disc;">... </ul>`
`circle`	○	`<ul style="list-style-type: circle;">... </ul>`
`square`	■	`<ul style="list-style-type: square;">... </ul>`
`none`	No bullet (plain list).	`<ul style="list-style-type: none;">... </ul>`

ORDERED LIST (<OL>)

- Ordered lists display items with numbers or letters by default.
  - <ol>
  - <li>Step 1</li>
  - <li>Step 2</li>
  - <li>Step 3</li>
  - </ol>

CHANGING THE NUMBERING STYLE

- The type attribute controls the style of the numbers or letters.
- Examples of type Attribute Values:

Value	Style	Example Code
`1` (default)	1, 2, 3...	`<ol type="1">...</ol>`
`A`	A, B, C...	`<ol type="A">...</ol>`
`a`	a, b, c...	`<ol type="a">...</ol>`
`I`	I, II, III...	`<ol type="I">...</ol>`
`i`	i, ii, iii...	`<ol type="i">...</ol>`

START ATTRIBUTE

- You can specify the starting value of an ordered list using the start attribute.
  - <ol start="5">
  - <li>Item 5</li>
  - <li>Item 6</li>
  - </ol>

NESTED LISTS

- Lists can be nested inside one another to create hierarchical structures.
  - <ul>
  - <li>Fruits
  - <ul>
  - <li>Apple</li>
  - <li>Banana</li>
  - </ul>
  - </li>
  - <li>Vegetables
  - <ul>
  - <li>Carrot</li>
  - <li>Broccoli</li>
  - </ul>
  - </li>
  - </ul>

DEFINITION LIST (<DL>)

- Definition lists present a set of terms and their corresponding descriptions.
  - <dl>
  - <dt>HTML</dt>
  - <dd>A markup language for structuring content on the web.</dd>
  - <dt>CSS</dt>
  - <dd>Used for styling HTML elements.</dd>
  - </dl>

STYLING LISTS WITH CSS

- You can customize lists using CSS for unique styles.
  - <style>
  - ul.custom-list {
  - list-style-type: square;
  - color: blue;
  - }
  - ul.custom-list li {
  - margin: 5px 0;
  - }
  - </style>
  - <ul class="custom-list">
  - <li>Item 1</li>
  - <li>Item 2</li>
  - <li>Item 3</li>
  - </ul>

BEST PRACTICES FOR USING LISTS

- Choose the Right List Type: Use unordered lists for unrelated items and ordered lists for steps or rankings.
- Avoid Over-Nesting: Too many nested lists can become confusing; keep it simple.
- Use Meaningful Descriptions: For accessibility, ensure <li> and <dd> text is clear and informative.

EMBEDDING MULTIMEDIA IN WEB PAGES

- Multimedia elements like images, audio, and video enhance the user experience by making web pages more engaging and interactive.
- HTML provides several tags to embed multimedia content directly into web pages.

EMBEDDING AUDIO IN HTML

- The <audio> tag is used to embed sound files in a webpage. It supports formats like MP3, Ogg, and WAV.
  - <audio controls>
  - <source src="audio.mp3" type="audio/mpeg">
  - <source src="audio.ogg" type="audio/ogg">
  - Your browser does not support the audio element.
  - </audio>

EMBEDDING AUDIO IN HTML

- `controls` **Attribute:** Adds play, pause, volume, and seek controls.
- `<source>` **Tag:** Specifies the audio file and format. Multiple `<source>` tags provide fallback options for different browsers.
- **Fallback Message:** Displays a message if the browser doesn't support the `<audio>` tag.

EMBEDDING AUDIO IN HTML

Attributes of `<audio>`:

Attribute	Description
`controls`	Displays built-in playback controls (play, pause, volume, etc.).
`autoplay`	Automatically starts playback when the page loads.
`loop`	Plays the audio in a continuous loop.
`muted`	Starts playback with the audio muted.

EMBEDDING VIDEO IN HTML

- The <video> tag is used to embed videos in a webpage. It supports formats like MP4, WebM, and Ogg.
  - <video controls width="640" height="360">
  - <source src="video.mp4" type="video/mp4">
  - <source src="video.ogg" type="video/ogg">
  - Your browser does not support the video element.
  - </video>

EMBEDDING VIDEO IN HTML

Explanation:

- `controls` **Attribute:** Adds playback controls for play, pause, volume, etc.
- `width` **and** `height` : Sets the dimensions of the video player.
- **Fallback Message:** Displays text if the browser doesn't support the `<video>` tag.

EMBEDDING VIDEO IN HTML

Attributes of `<video>`:

Attribute	Description
`controls`	Displays playback controls.
`autoplay`	Automatically plays the video when the page loads.
`loop`	Plays the video repeatedly in a loop.
`muted`	Starts video playback with no sound.
`poster`	Specifies an image to show while the video is loading or not playing.

```
<video controls poster="poster.jpg">
   <source src="movie.mp4" type="video/mp4">
</video>
```

EMBEDDING YOUTUBE VIDEOS

- You can embed YouTube videos using the <iframe> tag.
  - <iframe width="560" height="315" src="https://www.youtube.com/embed/VIDEO_ID"
  - title="YouTube video" frameborder="0" allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture"
  - allowfullscreen>
  - </iframe>

Explanation:

- Replace `VIDEO_ID` with the unique ID of the YouTube video.
- Use `allowfullscreen` for full-screen playback.

USING <EMBED> FOR OTHER MULTIMEDIA

- The <embed> tag can embed other multimedia types, like PDFs or Flash content.
  - <embed src="file.pdf" width="600" height="400" type="application/pdf">

USING <OBJECT> FOR COMPLEX EMBEDDING

- The <object> tag embeds multimedia elements and allows fallback content.
  - <object data="movie.mp4" type="video/mp4" width="640" height="360">
  - Your browser does not support this video.
  - </object>

BEST PRACTICES FOR EMBEDDING MULTIMEDIA

- Provide Fallbacks: Use alternative formats and clear messages for unsupported content.
- Optimize Media Files: Compress audio and video files for faster loading.
- Accessibility: Include captions, subtitles, or transcripts for media where applicable.
- Responsive Design: Ensure media scales properly using CSS for better display on various devices.
- Autoplay Use: Avoid autoplay on most web pages, as it can disrupt the user experience.

AUDIO, VIDEO, AND EMBED TOGETHER

```
<!DOCTYPE html>
<html>
<head>
  <title>Embedding Multimedia</title>
</head>
<body>
  <h1>Embedding Multimedia in Web Pages</h1>
    <h2>Audio Example</h2>
  <audio controls>
    <source src="audio.mp3" type="audio/mpeg">
    <source src="audio.ogg" type="audio/ogg">
    Your browser does not support the audio element.
  </audio>
```

AUDIO, VIDEO, AND EMBED TOGETHER

```
<h2>Video Example</h2>
  <video controls width="640" height="360" poster="poster.jpg">
    <source src="video.mp4" type="video/mp4">
    <source src="video.ogg" type="video/ogg">
    Your browser does not support the video element.
  </video>
    <h2>YouTube Video</h2>
  <iframe width="560" height="315" src="https://www.youtube.com/embed/dQw4w9WgXcQ"
        frameborder="0" allowfullscreen>
  </iframe>
    <h2>Embedding a PDF</h2>
  <embed src="file.pdf" width="600" height="400" type="application/pdf">
</body>
</html>
```

INSERTING TABLES IN HTML

- Tables are used in HTML to present data in rows and columns, making information organized and easy to read. The <table> tag and its associated tags structure and style tabular data.

```
<table>
    <tr>
        <th>Header 1</th>
        <th>Header 2</th>
        <th>Header 3</th>
    </tr>
    <tr>
        <td>Row 1, Column 1</td>
        <td>Row 1, Column 2</td>
        <td>Row 1, Column 3</td>
    </tr>
    <tr>
        <td>Row 2, Column 1</td>
        <td>Row 2, Column 2</td>
        <td>Row 2, Column 3</td>
    </tr>
</table>
```

TAGS USED IN TABLES

Tag	Description	Example Usage
<table>	Defines the table structure.	<table> ... </table>
<tr>	Defines a table row.	<tr> ... </tr>
<td>	Defines a data cell within a table row.	<td>Data</td>
<th>	Defines a header cell within a row, usually bold and centered by default.	<th>Header</th>
<caption>	Provides a title or description for the table (optional).	<caption>Table Caption</caption>
<thead>	Groups the header section of a table.	<thead><tr><th>Header</th></tr></thead>
<tbody>	Groups the body (data rows) of a table.	<tbody><tr><td>Data</td></tr></tbody>
<tfoot>	Groups the footer section of a table, useful for summaries.	<tfoot><tr><td>Footer</td></tr></tfoot>

ADDING BORDERS AND SPACING

```
<table border="1">
    <tr>
        <th>Column 1</th>
        <th>Column 2</th>
        <th>Column 3</th>
    </tr>
    <tr>
        <td>Row 1, Col 1</td>
        <td>Row 1, Col 2</td>
        <td>Row 1, Col 3</td>
    </tr>
</table>
```

CSS FOR BORDERS AND SPACING

```
<style>
  table {
    border-collapse: collapse; /* Removes default spacing between cells */
    width: 100%; /* Adjusts the table to the page width */
  }
  th, td {
    border: 1px solid black; /* Adds border to cells */
    text-align: center; /* Centers the text in the cells */
    padding: 8px; /* Adds padding to cells */
  }
</style>
```

MERGING CELLS

- You can use the colspan and rowspan attributes to merge cells horizontally or vertically.
  - <table border="1">
  - <tr>
  - <th colspan="3">Merged Header</th>
  - </tr>
  - <tr>
  - <td>Row 1, Col 1</td>
  - <td>Row 1, Col 2</td>
  - <td>Row 1, Col 3</td>
  - </tr>
  - </table>

MERGING CELLS

- Vertical Cell Merge with rowspan
  - <table border="1">
  - <tr>
  - <td rowspan="2">Merged Row</td>
  - <td>Row 1, Col 2</td>
  - </tr>
  - <tr>
  - <td>Row 2, Col 2</td>
  - </tr>
  - </table>

ADDING A TABLE CAPTION

```
<table border="1">
    <caption>Student Grades</caption>
    <tr>
        <th>Student Name</th>
        <th>Subject</th>
        <th>Grade</th>
    </tr>
    <tr>
        <td>John Doe</td>
        <td>Math</td>
        <td>A</td>
    </tr>
</table>
```

GROUPING TABLE SECTIONS

- Grouping rows into <thead>, <tbody>, and <tfoot> improves readability and styling control.
  - <table border="1">
  - <thead>
  - <tr>
  - <th>Student Name</th>
  - <th>Subject</th>
  - <th>Grade</th>
  - </tr>
  - </thead>

```
<tbody>
    <tr>
        <td>John Doe</td>
        <td>Math</td>
        <td>A</td>
    </tr>
    <tr>
        <td>Jane Smith</td>
        <td>English</td>
        <td>B+</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
        <td colspan="3">End of Table</td>
    </tr>
  </tfoot>
</table>
```

STYLING TABLES WITH CSS

```
<style>
    table {
        border-collapse: collapse;
        width: 100%;
    }
    th {
        background-color: #f4f4f4;
        font-weight: bold;
    }
th, td {
    border: 1px solid #ddd;
    text-align: left;
    padding: 10px;
  }
  tr:nth-child(even) {
    background-color: #f9f9f9;
  }
</style>
```

STYLING TABLES WITH CSS

```
<table>
    <tr>
        <th>Header 1</th>
        <th>Header 2</th>
        <th>Header 3</th>
    </tr>
    <tr>
        <td>Row 1, Col 1</td>
        <td>Row 1, Col 2</td>
        <td>Row 1, Col 3</td>
    </tr>
    <tr>
        <td>Row 2, Col 1</td>
        <td>Row 2, Col 2</td>
        <td>Row 2, Col 3</td>
    </tr>
</table>
```

BEST PRACTICES

- Use Descriptive Captions: Provide meaningful titles with <caption>.
- Style with CSS: Avoid using inline styles for complex tables; rely on CSS classes.
- Accessibility: Use <th> for headers and the scope attribute to improve readability for assistive technologies.Example: <th scope="row">Header</th>

INTERNAL LINKING IN HTML

- Internal links connect to other pages or sections within the same website or document. They help users navigate efficiently within a site.
- Basic Syntax for Internal Links
  - <a href="page.html">Visit Page</a>
- Linking to Another Page in the Same Folder
  - <a href="about.html">About Us</a>
- Linking to a Page in a Subfolder
  - <a href="subfolder/contact.html">Contact Us</a>
- Relative Path to Navigate Up a Directory
  - <a href="../services.html">Our Services</a>
- Linking to a Specific Section within a Page
  - <!-- Target Section -->
  - <h2 id="section1">Section 1</h2>

  - <!-- Link to Target Section -->
  - <a href="#section1">Go to Section 1</a>

EXTERNAL LINKING

- External links direct users to a webpage on another website, typically using a full URL.
- Basic Syntax for External Links
  - <a href="https://example.com">Visit External Site</a>
- Simple External Link
  - <a href="https://www.google.com">Visit Google</a>
- Opening Links in a New Tab
  - <a href="https://www.wikipedia.org" target="_blank">Visit Wikipedia</a>
- Adding a Tooltip (Optional)
  - <a href="https://www.github.com" title="Go to GitHub">GitHub</a>

COMBINING INTERNAL AND EXTERNAL LINKS

```
<!DOCTYPE html>
<html>
<head>
    <title>Links Example</title>
</head>
<body>
    <h1>Welcome to My Website</h1>
    <!-- Internal Link -->
    <a href="about.html">About Us</a>

    <!-- External Link -->
    <a href="https://example.com" target="_blank">Visit Example</a>

<!-- Internal Link to a Section -->
   <a href="#contact">Go to Contact Section</a>

   <!-- Content to Scroll -->
   <div style="margin-top: 800px;" id="contact">
      <h2>Contact Us</h2>
      <p>Email: contact@example.com</p>
   </div>
</body>
</html>
```

BEST PRACTICES FOR LINKING

- Descriptive Link Text: Use meaningful text for links rather than generic terms like "Click here.“
  - Example: Use <a href="contact.html">Contact Us</a> instead of <a href="contact.html">Click here</a>.
- Accessible Links: Include tooltips (title) to provide more context.
  - Example: <a href="https://example.com" title="Visit Example.com">Example</a>.
- Opening in New Tabs: Use target="_blank" only for external links or resources that users shouldn’t navigate away from (e.g., terms of service).
- Consistency in Relative Paths: Use a consistent structure for internal links to prevent navigation errors.

HTML FRAMES

- HTML frames were originally used to divide a webpage into multiple sections, each displaying a different document or part of a page.
- However, **frames are outdated and no longer recommended for modern web development**, as they were replaced by more efficient technologies like CSS and JavaScript.
- Despite their deprecation in HTML5, they are still worth understanding for legacy systems.

FRAMESET EXAMPLE

- A **frameset** is used to divide the browser window into multiple sections.
  - <!DOCTYPE html>
  - <html>
  - <head>
  - <title>HTML Frames Example</title>
  - </head>
  - <frameset cols="50%,50%">
  - <frame src="page1.html">
  - <frame src="page2.html">
  - </frameset>
  - </html>

Explanation:

- `cols="50%,50%"` : Divides the browser window into two vertical sections, each taking 50% of the width.
- `<frame src="page1.html">` : Embeds `page1.html` in the first frame.
- `<frame src="page2.html">` : Embeds `page2.html` in the second frame.

ROW DIVISION WITH FRAMES

- You can divide the page horizontally using rows.
  - <frameset rows="70%,30%">
  - <frame src="header.html">
  - <frame src="footer.html">
  - </frameset>

  - Output:Top 70% of the window shows header.html.Bottom 30% of the window shows footer.html.

NESTED FRAMESETS

- Framesets can be nested to create more complex layouts.
  - <frameset rows="20%,80%">
  - <frame src="header.html">
  - <frameset cols="60%,40%">
  - <frame src="content.html">
  - <frame src="sidebar.html">
  - </frameset>
  - </frameset>

INLINE FRAMES (IFRAME)

- Unlike <frame>, the <iframe> tag is widely used in modern web design to embed content such as videos, maps, or external websites.
  - <!DOCTYPE html>
  - <html>
  - <head>
  - <title>iFrame Example</title>
  - </head>
  - <body>
  - <h1>iFrame Example</h1>
  - <iframe src="https://www.example.com" width="600" height="400" frameborder="0"></iframe>
  - </body>
  - </html>

ATTRIBUTES OF <IFRAME>

Attribute	Description
src	Specifies the source of the content to be embedded.
width and height	Define the size of the iframe in pixels.
frameborder	Sets the thickness of the iframe border (default: 1; 0 removes the border).
allowfullscreen	Allows the iframe to be viewed in full-screen mode.

ADVANTAGES AND DISADVANTAGES OF FRAMES

- **Advantages:**
  - **Consistent Layouts:** Allows static elements (e.g., menus) while the rest of the page changes.
  - **Content Segregation:** Each section of a frame is independent.
  - **Simplifies Updates:** Replacing content in a frame updates only that section.
- **Disadvantages:**
  - **Deprecated in HTML5:** Frames are obsolete and no longer supported by modern browsers.
  - **Poor SEO Performance:** Search engines struggle to index frame-based content.
  - **User Experience Issues:** Links may not function as expected, and navigation becomes complex.
  - **Incompatibility:** Not suitable for responsive design or small screens.

ALTERNATIVES TO FRAMES

- Use **CSS Flexbox** or **Grid** for dividing and managing sections.
  - .container {
  - display: grid;
  - grid-template-columns: 1fr 3fr;
  - }

- JavaScript or AJAX
  - Fetch and load parts of content dynamically without reloading the entire page.

- Include Using Server-Side Scripting
  - Use technologies like PHP or ASP.NET to include reusable headers, footers, or menus.

HTML FORMS

- Forms in HTML are used to collect user input and submit it to a server for processing. HTML provides various form elements to capture data, such as text fields, dropdowns, radio buttons, and checkboxes.
- The <form> tag creates a container for all form elements.
  - <form action="submit.php" method="post">
  - <!-- Form elements go here -->
  - </form>

KEY ATTRIBUTES OF <FORM>

Attribute	Description	Example
action	Specifies the URL where the form data will be submitted.	action="submit.php"
method	Specifies HTTP method: GET (default) or POST.	method="post"
enctype	Specifies how form data is encoded (e.g., application/x-www-form-urlencoded or multipart/form-data).	enctype="multipart/form-data"

FORM ELEMENTS

Text Input Fields

- Single-line Text Input:
  - <input type="text" name="username" placeholder="Enter your name">
- Password Input:
  - <input type="password" name="password" placeholder="Enter your password">
- Multi-line Text Input (Text Area):
  - <textarea name="comments" rows="4" cols="50" placeholder="Enter your comments"></textarea>

FORM ELEMENTS

- Radio Buttons (Select One Option):
  - <label><input type="radio" name="gender" value="male"> Male</label>
  - <label><input type="radio" name="gender" value="female"> Female</label>

- Checkboxes (Select Multiple Options):
  - <label><input type="checkbox" name="skills" value="HTML"> HTML</label>
  - <label><input type="checkbox" name="skills" value="CSS"> CSS</label>
  - <label><input type="checkbox" name="skills" value="JavaScript"> JavaScript</label>

FORM ELEMENTS

Dropdown List

- Single Selection:
  - <select name="country">
  - <option value="usa">USA</option>
  - <option value="uk">UK</option>
  - <option value="india">India</option>
  - </select>

- Multiple Selections:
  - <select name="hobbies" multiple size="3">
  - <option value="reading">Reading</option>
  - <option value="traveling">Traveling</option>
  - <option value="coding">Coding</option>
  - </select>

FORM ELEMENTS

Buttons

- Submit Button:
  - <button type="submit">Submit</button>

- Reset Button:
  - <button type="reset">Reset</button>

FORM ELEMENTS

- File Upload Field:
  - <input type="file" name="resume">

- Hidden Field (Invisible to Users):
  - <input type="hidden" name="user_id" value="12345">

VALIDATION ATTRIBUTES

- HTML provides built-in form validation attributes for better user experience.

Attribute	**Description**	**Example**
required	Ensures the field cannot be left blank.	<input type="text" required>
maxlength	Sets the maximum number of characters allowed.	<input type="text" maxlength="10">
min and max	Define numeric value limits.	<input type="number" min="1" max="100">
pattern	Ensures input matches a specified regular expression.	<input type="text" pattern="[A-Za-z]+">
placeholder	Displays a short hint inside the input field.	<input type="text" placeholder="Hint">

EXAMPLE: A SIMPLE HTML FORM

```
<!DOCTYPE html>
<html>
<head>
    <title>HTML Forms</title>
</head>
<body>
    <h1>Registration Form</h1>
    <form action="/submit.php" method="post">
        <label for="username">Name:</label>
        <input type="text" id="username" name="username" required placeholder="Enter your name">
        <br><br>
```

EXAMPLE: A SIMPLE HTML FORM

- <label for="password">Password:</label>
- <input type="password" id="password" name="password" required>
-

- <label>Gender:</label>
- <label><input type="radio" name="gender" value="male"> Male</label>
- <label><input type="radio" name="gender" value="female"> Female</label>
-

EXAMPLE: A SIMPLE HTML FORM

```
<label for="country">Country:</label>
    <select id="country" name="country" required>
        <option value="">Select</option>
        <option value="usa">USA</option>
        <option value="uk">UK</option>
        <option value="india">India</option>
    </select>
    <br><br>
```

EXAMPLE: A SIMPLE HTML FORM

- <label>Skills:</label>
- <label><input type="checkbox" name="skills" value="HTML">
HTML</label>
- <label><input type="checkbox" name="skills" value="CSS"> CSS</label>
- <label><input type="checkbox" name="skills" value="JavaScript">
JavaScript</label>
-

- <label for="file">Upload Resume:</label>
- <input type="file" id="file" name="resume" accept=".pdf,.doc">
-

EXAMPLE: A SIMPLE HTML FORM

- <button type="submit">Submit</button>
- <button type="reset">Reset</button>
- </form>
- </body>
- </html>

BEST PRACTICES FOR FORMS

- Use Proper Validation: Implement both client-side (HTML5) and server-side validation.
- Accessibility: Use labels and for attributes for better screen reader support.
- Placeholder vs Label: Always provide labels; placeholders should only be hints.
- Security: Protect form data with HTTPS and sanitize inputs on the server to prevent SQL injection and XSS.